The intricacies of designing a recurrence calendar

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LU GAN December 4, 2021

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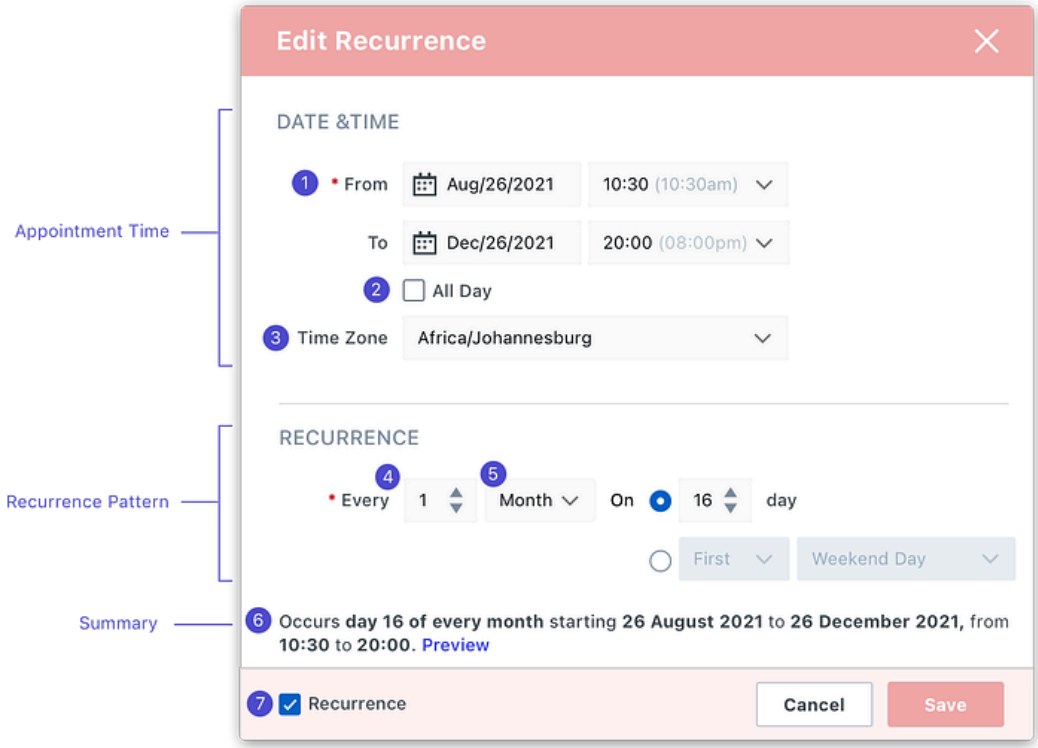
Though the function of setting recurrence is a good design, tweaking it has always been a bulk of work to me. When I opened a recurrence page or popup window, I just felt a lot of information and options were thrown on my face and I had to deal with every single setting deliberately to avoid a single mistake.

Similarly, designers who create a recurrence calendar may also feel overwhelmed in the first place. They can come up with as many options as possible that pertain to users' needs; yet, in the actual design process, they must narrow the scope of options so as to not confuse and frustrate users.

In the nextsection, I will quickly give you a holistic view of the recurrence calendar and walk you through its basic components. Besides, some of my suggestions will be presented. If you want to know more about the complexity of recurrence calendar design or dig deep into how to design a better calendar to meet users' needs, just keep scrolling down to the section.

I. Basic Components

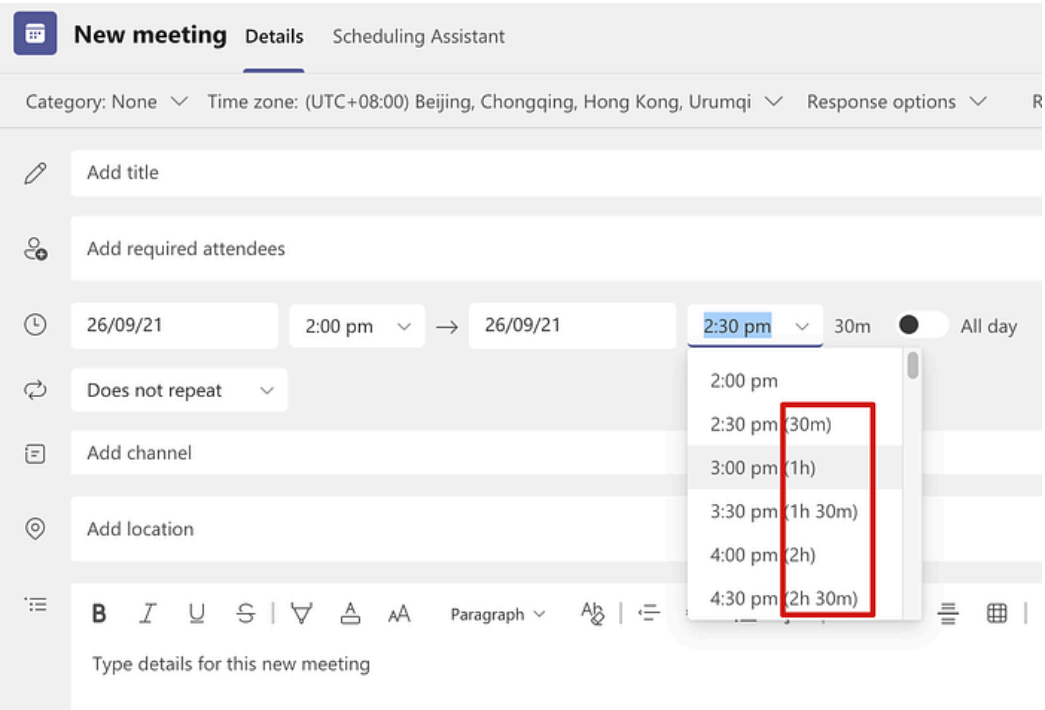
A recurrence calendar can be divided into three parts: *Appointment Time*, *Recurrence Pattern,* and *Summary*



1. Date & Time: *Use to set when to start and end a certain event recurrence*

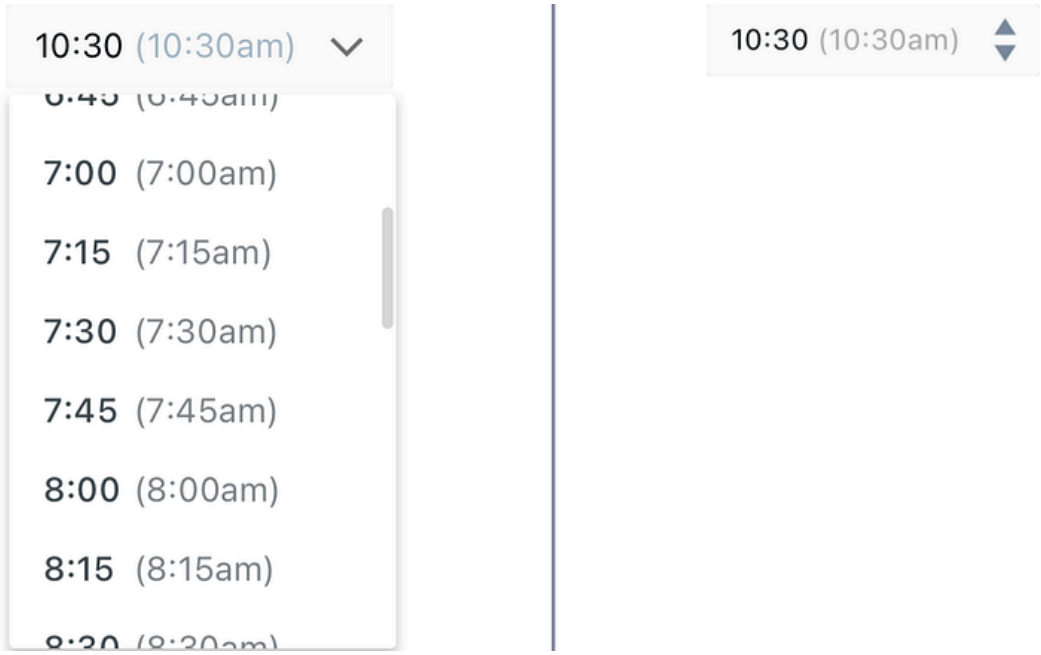
Time format: I notice a lot of designs for time setting provide *am* and *pm* options, but a *24 hour-time* format should be also considered — it is rather familiar for European users.

To make time setting more flexible and easy to use, I employ the 24hour-time format while putting am/pm format side by side as a hint (see picture above). If users reveal no concern about the time format, then putting the duration time as a hint on the end time would be a good choice.



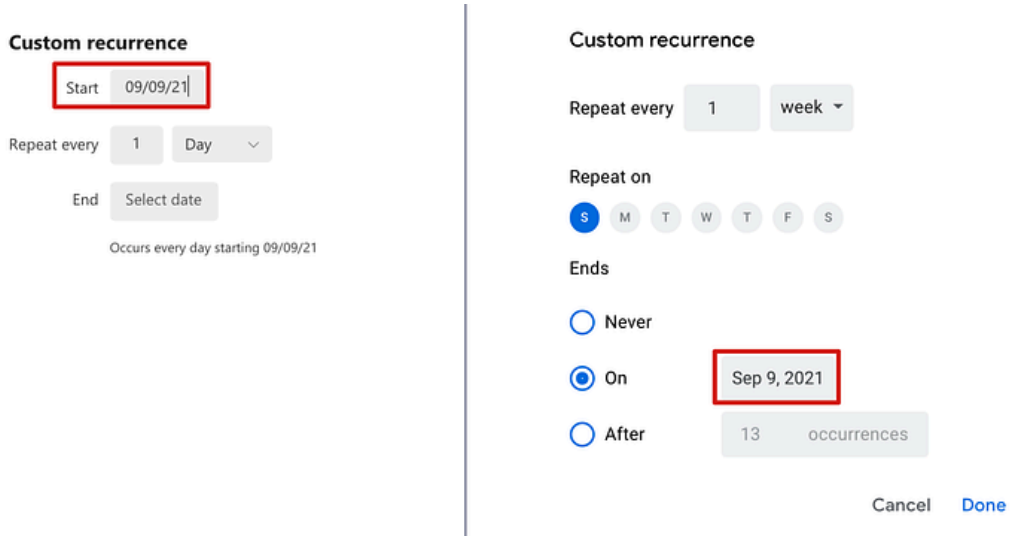
😊 displaying the duration for users to reassure.

In addition, providing users with a bunch of pre-defined time slots in a dropdown menu would help users avoid some laborious work: neither switching back and forth between mouse and keyboard nor constantly clicking the up and down arrows is a good user experience.



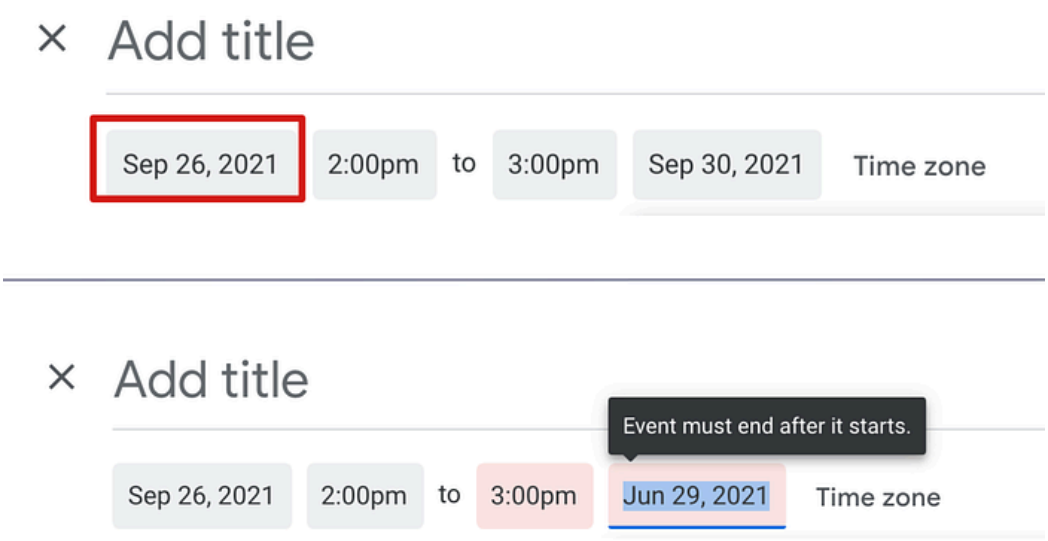
😊 Users can only depend on a mouse to complete their tasks. 😣 User has to either constantly click up and down arrows, or use a keyboard to input time.

Start & End Date: Keep in mind what date format your users prefer: in some countries, the date format is *YYYY-MM-DD*, others are *MM-DD-YYYY or DD-MM-YYYY, etc.* Hence, be sure to use the right format. Alternatively, you can use an **abbreviated month** or **spell out the month** ratherthanshowing numbers to let users easily tell months and days.



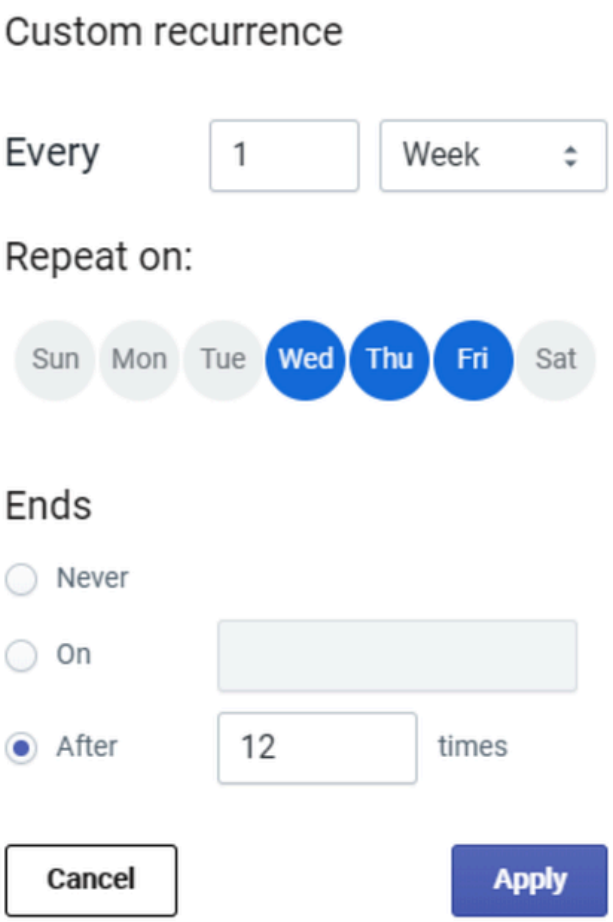
😣 Hard to tell which one indicates the month and which one is the day(). 😊 : Month and day can be easily distinguished (*Google Calendar*)

The other thing that draws attention is that the end date should not be earlier than the start date. It may sound like common sense, though I still see some designs make such a mistake.



: Users can select an end date earlier than the start date. Though the system will notify users that the end date they are selecting is incorrect, this non-error prevention design is not user-friendly. 😣 It is very annoying to see the error notification and tooltips only after users made a mistake. Why couldn't limit the range of the date at the beginning? *Google Calendar*

End time: As you may also notice, a lot of calendars feature the "*After XXX Occurrences*" option on the End time. This function can help users finish their tasks quickly without calculating when will be the date to end a certain event if they already know how many repeats will occur. Besides, *No end time* option should be included in the End time setting.

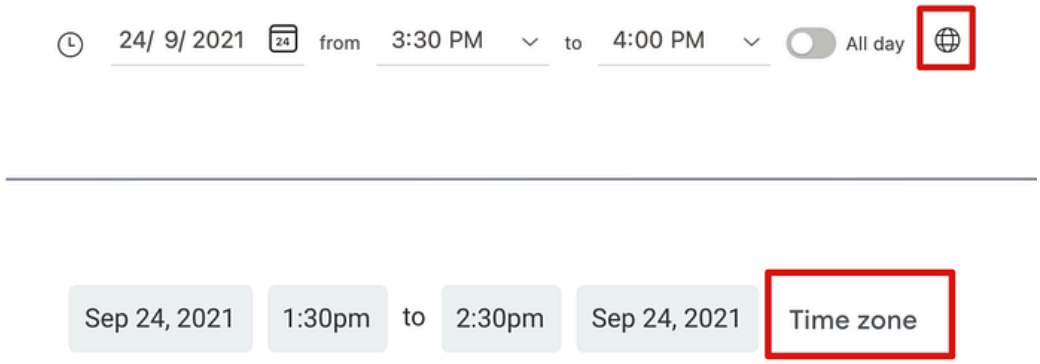


2. All Day: *If this one has been selected, it means an event will be held without a specific duration, hence, the start time and end time should be disabled accordingly.*

3. Time Zone: *Use to select which time zone user wants to apply*

There are few recurrence calendars using an icon to display time zone, like Outlook. It can definitely save more space and is less distractive for users to complete their task, but it also hides and users may easily forget to choose the right time zone.

Therefore, placing a full setting or just presenting an icon really depends on users' mental modal, how frequently this setting will be used and how designers want to guide their users.



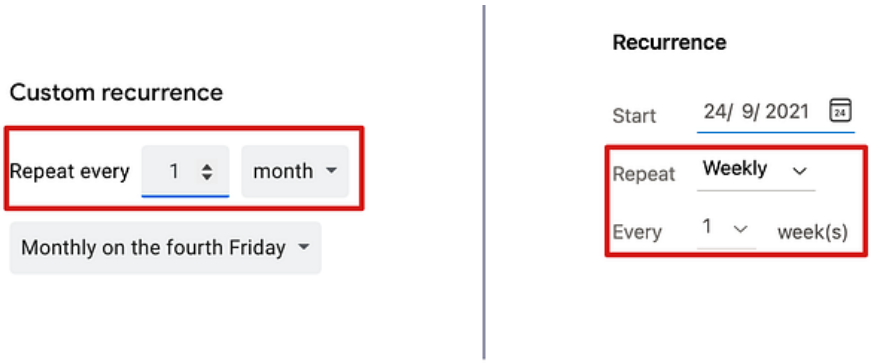
uses an icon to display time zone. [Google Calendar](#) uses text to show time zone.

4. Recurrence Value: *use to define how often users want to recurrent*

In designing this setting, it is important to ensure to set a default value as 1. This is for error prevention. I noticed some designs had left this value input empty. Once users hit the save button, an error message would pop up to force value input on the user side. With default value, users would not have this type of issue and get their tasks done smoothly.

5. Recurrence Unit: *recurring appointments can occur every day (Daily), once a week(Weekly), once a month (Monthly), or once a year (Yearly).*

There are two different designs about the setting, some designs combine recurrence value and unit as one group, others separate them.

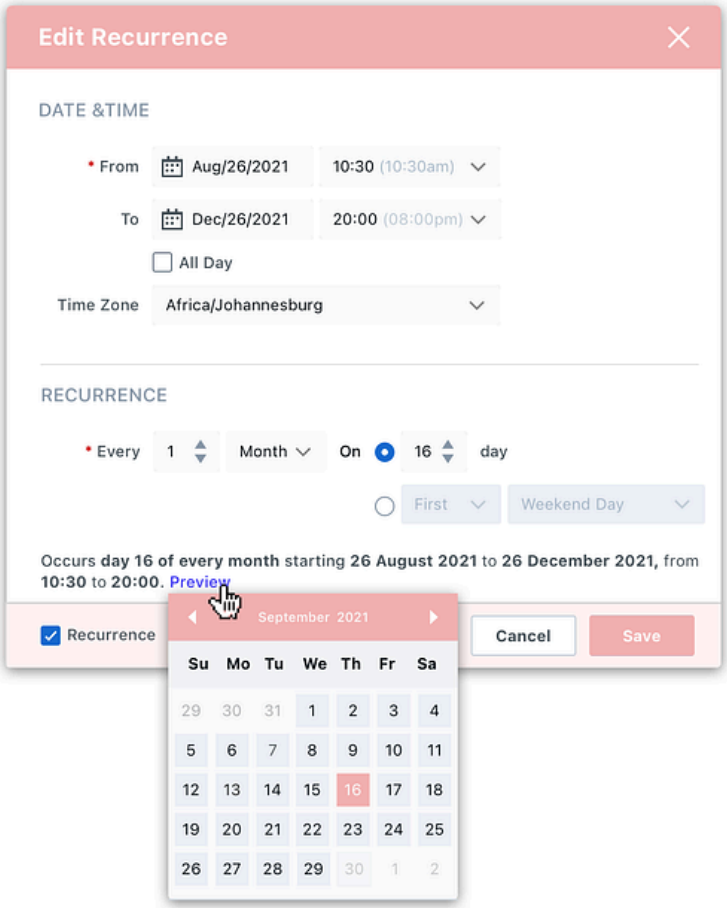


Combine Recurrence value and unit([Google Calendar](#)). : Place Recurrence value and unit in two rows ().

6. Summary Text: *it can give users a chance to check whether they select the right date, time, and recurrences schedule.*

The text itself should include most of the settings such as start and end dates and time, as well as the frequency of recurrence.

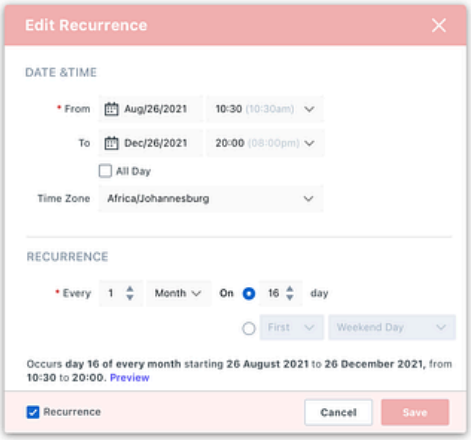
The *preview* function is optional, it is nice to give users a visual look to reassure their selections. However, due to limited time and dev resources in the real world, this feature can be excluded. Below is the preview function:

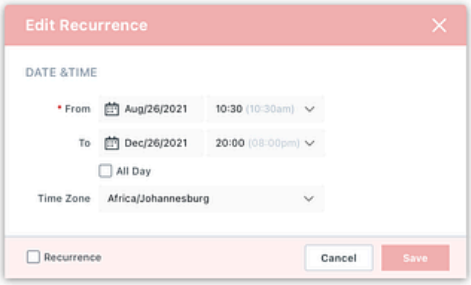


When users hover over the preview text, the date that users choose to repeat will be highlighted.

7. Recurrence checkbox: decide whether making a given event/appointment recurrent

It is important to set a default value for it during the design process. Check the box by default if most users would like to repeat or vice versa.





: Recurrence has been selected by default. : Recurrence has been unselected by default.

II. Complexity and Users' Goals

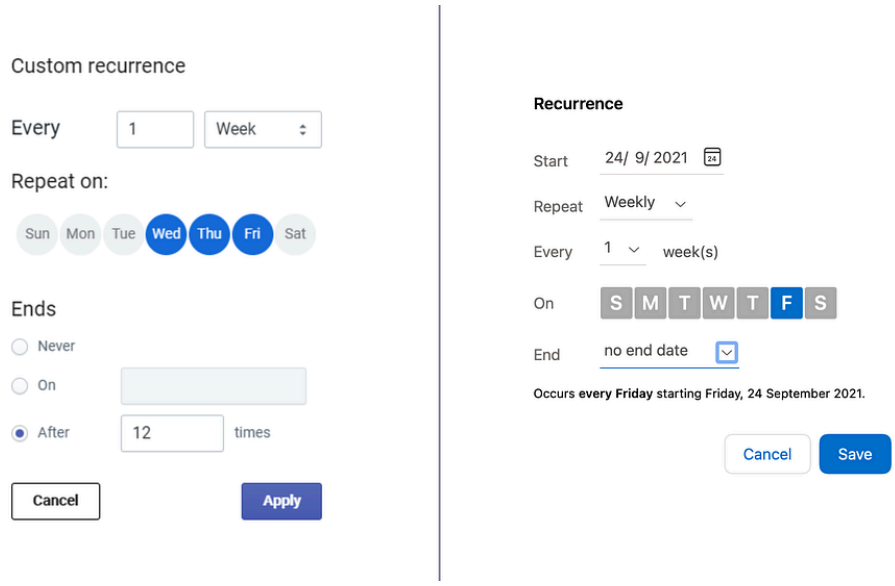
📅 Daily-basis

It is a pretty simple design and I will not elaborate on it in this article

📅 Weekly-basis

Here are several designs I want to share, the first one is from Outlook that uses the first letter of each day of the week to display. By contrast, Google calendar uses the first three letters of each day of the week to present.

The pros for Outlook calendar are obvious, it takes less space and provides less visual noise for users. Meanwhile, the cons may not be palpable to users at the first glance but could turn alarming at the globalization stage as users in certain countries/regions prefer two or even the whole letters to indicate each day of the week. So it is crucial to make the design decision in line with different groups of users.



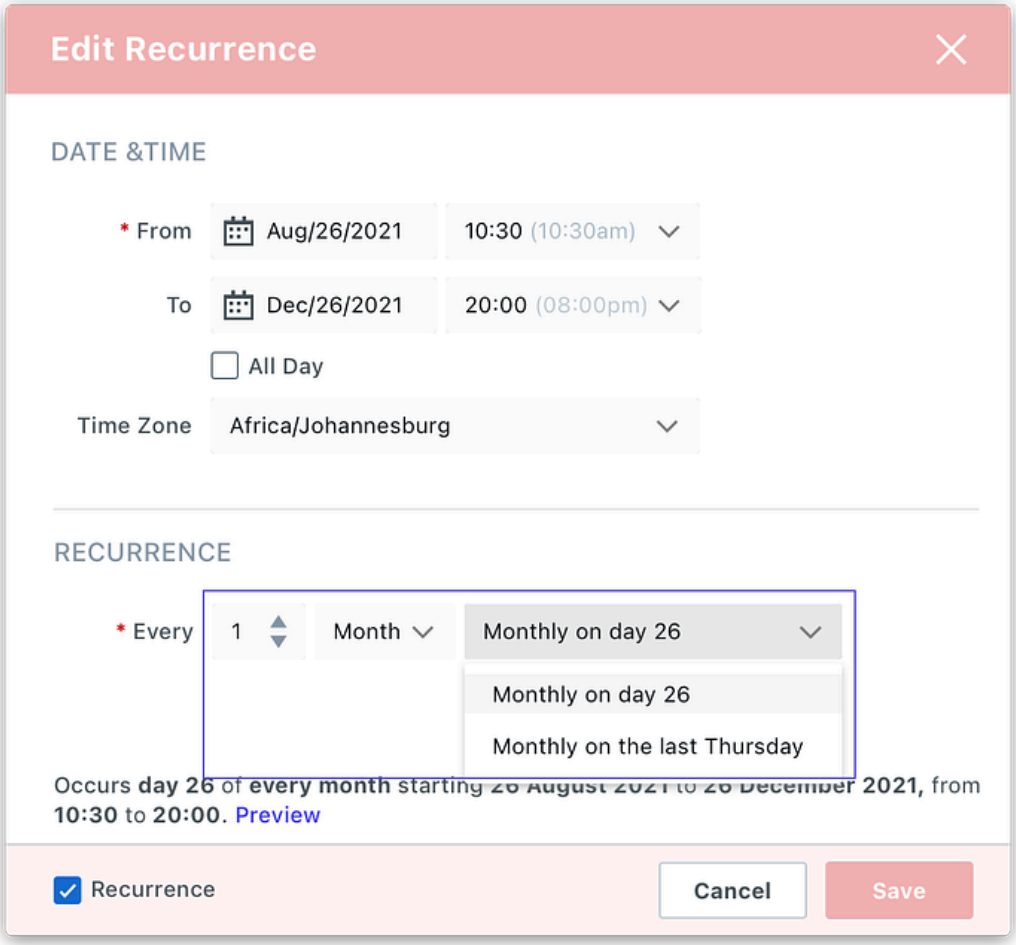
use the first three letters of each day of the week to present (Source: *Google Calendar*). : use the first letter of each day of the week to present (Source:).

📅 Monthly-basis

It is more complicated than a weekly-basis setting. UI for monthly-basis can vary by the goals users aim to achieve.

Below are three levels that designers can choose given different user goals.

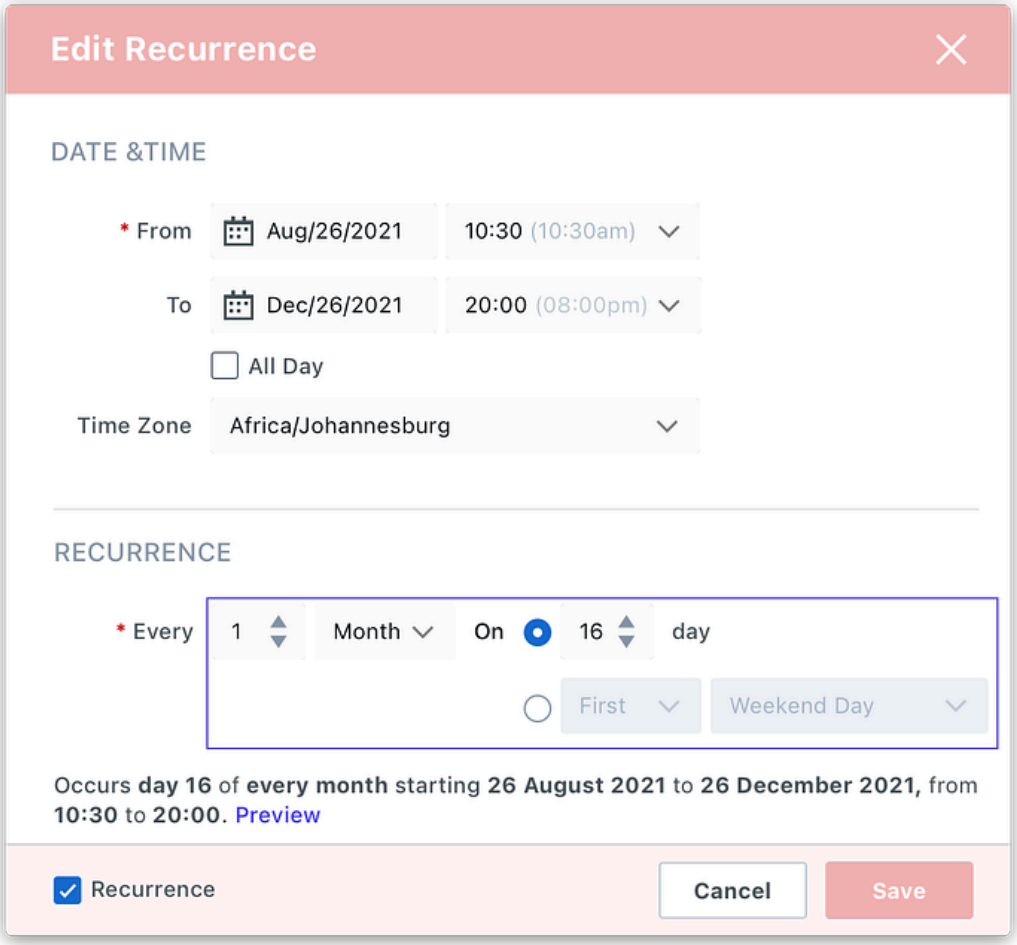
: can repeat only ONCE a month WITH specific date selected



Users can only select where it should repeat on the same date (e.g. monthly on day 26) or the day's week of this month and the day of this week (e.g. monthly on the last Thursday). — *Google Calendar* is using this recurrence pattern.

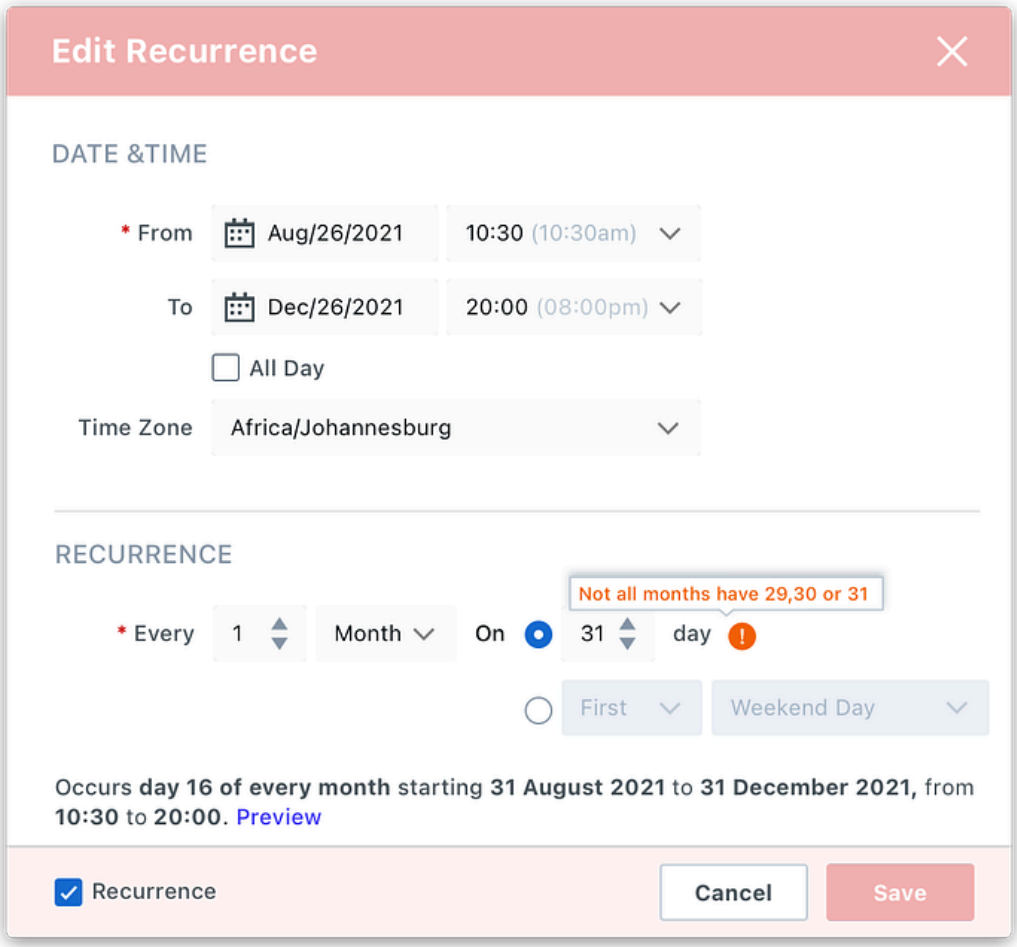
> **Goals user can achieve:**
>
> Repeat event/appointment monthly on **today's date** of this month;
>
> Repeat event/appointment monthly on **today's week of this month and today's day of this week**.

repeat only ONCE a month WITHOUT specific date selected

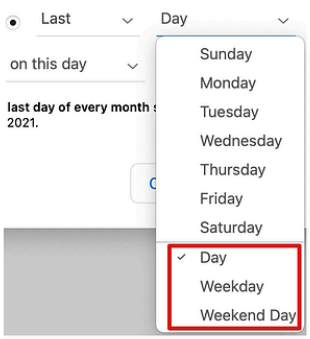


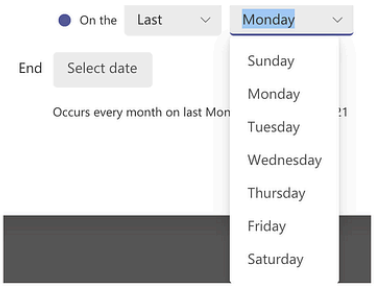
andare using this recurrence pattern

> **Goals user can achieve:**
>
> Repeat event/appointment **monthly** on **any date** of a month;
>
> Repeat event/appointment **monthly** on **any week of the month and day of the week.** (such as ‘On First Weekend Day’ of every month)
>
> ***Pay attention to…**
>
> Not every month has a **31st**, and in some years, February could have a **29th**. It is better to place a warning info icon (or whatever you think can better help users to recognize the fact) where a warning text would pop up upon users hovering over it(see below).



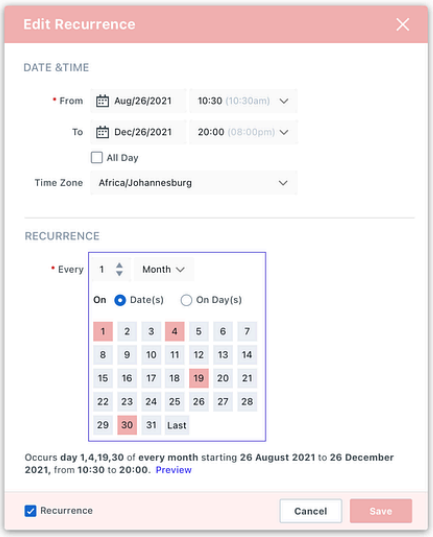
> It may not seem clear to select **Last Sunday**, it can be hard for users to estimate when their events/appointments will occur. Besides, if users want to set the last day of every month as recurrent time, how can he/she achieve the goal? It would be nice to have **Day**, **Weekday** and **Weekend Day** (see below).

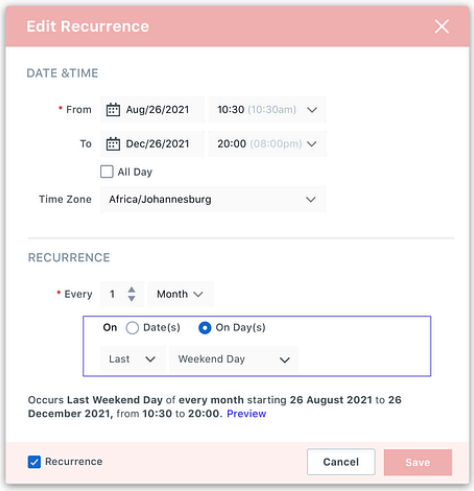




😊 Outlook Calendar features Day, Weekday, and Weekend Day to give users more control. 😣 : Microsoft Teams does not have Day, Weekday, and Weekend Day. can be vague for users.

: can repeat as MANY dates as users prefer per month





Users can select as many dates as they want to repeat for every month, Route4Me is using this recurrence pattern.

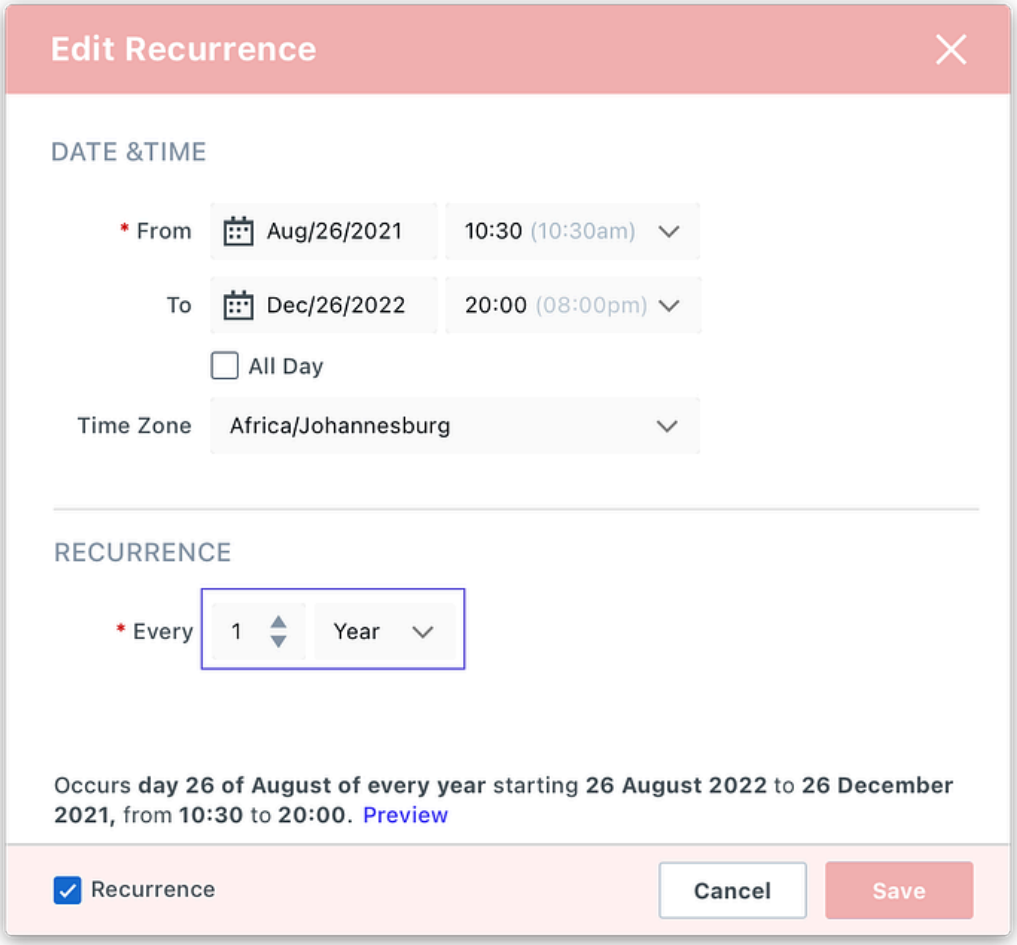
> **Goals user can achieve:**
>
> Repeat event/appointment **monthly** on **any date(s)**;
>
> Repeat event/appointment **monthly** on **any week of the month and day of the week**(such as ‘On First Monday’ of every month).
>
> ***Pay attention to…**
>
> Still, when designing the third level recurrence page, designers should add **Last** option on the date picker because some months **do not have a 31st** and February could have a **29th**.

📅 Yearly-basis

Very similar to monthly-basis design logic, yearly-basis design also needs to pay attention to several places and has three different ways to design.

Below are three levels that designers can choose given different user goals.

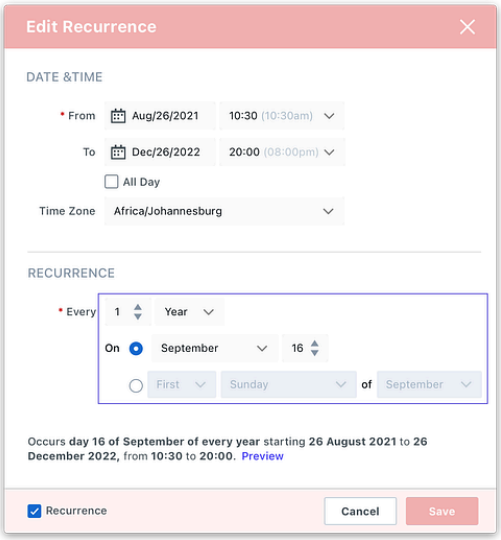
can repeat only ONCE a year on SETTING DAY’S DATE

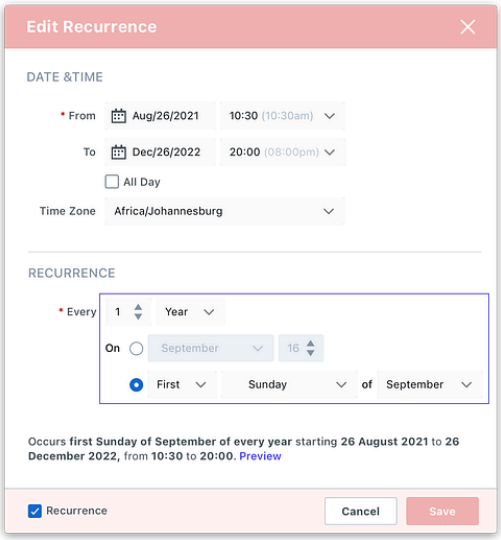


Can only repeat on the same date and month every year [Google Calendar](#)is using this recurrence pattern.

> **Goal users can achieve:**
>
> Repeat event/appointment **yearly** on **today's month and date**.

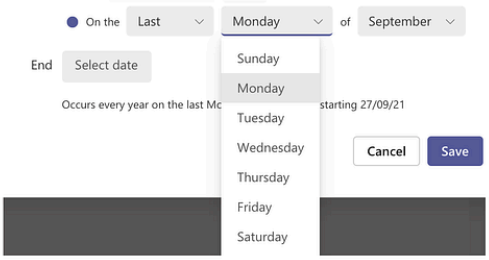
repeat only ONCE a year in ANY given month

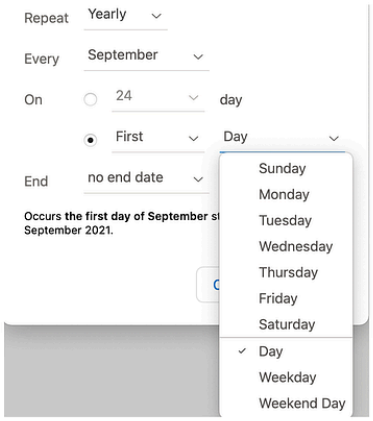




: user can select any month and any date to repeat. : user can select any day of a week and month to repeat (Microsoft Teamsandare using this recurrence pattern

> **Goals users can achieve:**
>
> Repeat event/appointment on certain DD/MM each year or every a couple of years
>
> Repeat event/appointment **yearly** on **any week of the month, any day of the week of any month**(such as 'On First Sunday of September' of every year).
>
> ***Pay attention to…**
>
> *Still, show , , and on the dropdown menu will be much helpful for users to understand.(see below).*

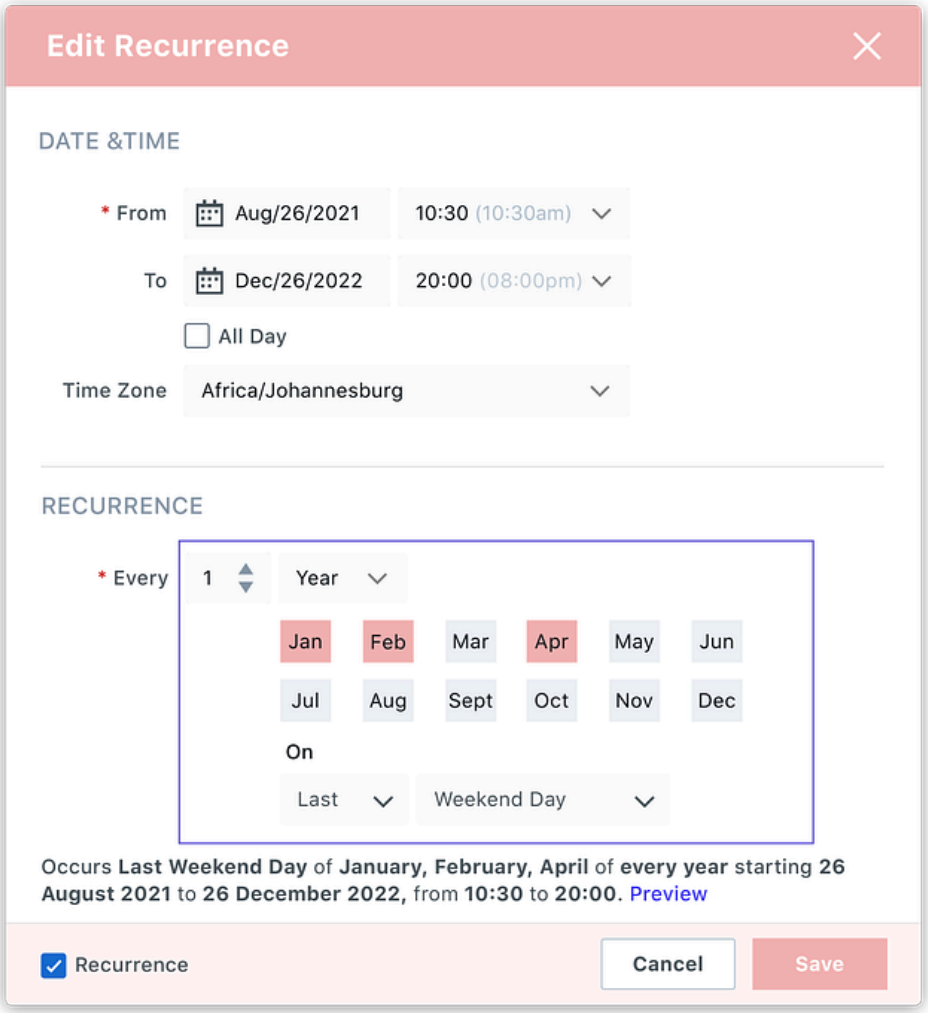




does not have Day, Weekday, and Weekend Day. 😊 : features Day, Weekday, and Weekend Day, which makes it easier to understand.

> *Also, not every year has the of February. If user has been setting recurrence event/appointment yearly on February 29th, system should remind users when they have selected this specific date.*

: repeat as MANY months as users prefer per year



Users can select as many months as they want to repeat for every year, Route4Me is using this recurrence pattern.

> **Goal users can achieve:**
>
> Repeat event/appointment on certain day (e.g., first, second) in certain month(s) each year or every a couple of years.
>
> ***Pay attention to…**
>
> During the stage of globalisation, the space of month container box should be set wider enough. Because the abbreviation length of month can vary among different countries: some of them may only have 3 letters while others can have 5 or even more letters. Know more

Conclusion

Of course, there are a lot of other existing designs of recurrence. Designers should try really hard to realize what users care about and what goals they are trying to achieve. Recurrence calendar is a task-heavy page that designers need to meticulously design and arrange all these functions. Considering work limits such as time, financial, and dev resources, designers should know which functions are *must-have*, which are *nice to have* or *not needed* (MVP).

Takeaway

1. Be sure to use the user-preferred to present the Date and Time.
2. Showing the of an event.
3. Offering a bunch of time slots.
4. Use an orrather than showing numbers.
5. option should be included in the End time setting.
6. Any event's end date should its start date.
7. Provide users with for recurrence value.
8. is a nice-to-have function.
9. Choose wisely between an abbreviation and full display when showing .
10. Provide different recurrence settings based on your .
11. Add a option on the date picker.
12. It would be nice to have , , and to fine-tune your recurrence pattern.
13. Choose wisely when displaying , and think about how much design space each month needs to take.

References

Using the Recurrence Pattern Dialog Box

The recurrence pattern dialog box is a simple way to set up complex recurring appointments. To open the dialog box…

www.infragistics.com

Create Master Route & Create Recurring Schedule

Route4Me's Scheduled & Recurring Routing is designed for all types of last mile businesses who want to automate…

support.route4me.com